

Contents

1 JW_HVB Expansion Port — Hardware Iteration Reference	1
1.1 1. Scope	1
1.2 2. Power Analysis	1
1.3 3. Currently Consumed Pins	2
1.4 4. Available Pins	3
1.5 5. Solutions Comparison	3
1.6 6. Recommended Configuration (Solution 1)	9
1.7 7. Hardware Design Details	11
1.8 8. Expansion Connector Specification	18
1.9 9. Fallback Pin Options	19
1.10 10. Firmware Impact	20
1.11 11. Decision Rationale	21
1.12 12. Open Questions for PCB Designer	21

1 JW_HVB Expansion Port — Hardware Iteration Reference

Generated: 2026-07-01 Based on: ref/pin_map.md, ref/board_design.md, boards/jianwei/jw_hvb/jw_hv

1.1 1. Scope

Add an expansion port to the jw_hvb board hosting:

Component	Model	Interface	Notes
Environmental sensor	BME280	I2C (addr 0x76/0x77)	Temperature, humidity, pressure
OLED display	Waveshare SSD1331	SPI (write-only)	96×64 RGB OLED, 262K color
Rotary encoder	generic quadrature	2× GPIO + 1× SW	With push-button
Buttons	4× tactile	GPIO inputs	User input

1.2 2. Power Analysis

1.2.1 2.1 Existing Rail: VCC_+3V3 (U28 MP1470GJ, 3A buck)

Consumer	Typ (mA)	Max (mA)
STM32F429BIT6 @168 MHz	100	200
SHT31A (I2C1, addr 0x44)	0.6	3.4
MAX3485 RS-485 transceiver	1	2
CMT8261N1 isolators (4×, MCU side)	20	40
Pull-ups, LED, misc	5	15
Existing subtotal	~127	~260
BME280 (measuring)	0.5	3.4

Consumer	Typ (mA)	Max (mA)
SSD1331 OLED (typ brightness)	25	50
Rotary encoder pull-ups	<0.5	<1
4× button pull-ups	<1	<2
New subtotal	~27	~56
Grand total	~154	~317

1.2.2 2.2 Voltage Compatibility

Component	Vdd range	3.3 V compatible?
BME280	1.71–3.6 V	Yes
SSD1331 module (Waveshare)	3.3 V / 5 V	Yes (3.3 V version)
Rotary encoder	passive	Yes
Tactile buttons	passive	Yes

Verdict: No modification to power architecture needed. VCC_+3V3 has ~2.7 A headroom. All devices operate at 3.3 V logic levels.

1.3 3. Currently Consumed Pins

Extracted from boards/jianwei/jw_hwb/jw_hvb.dts and ref/pin_map.md.

```

PA04  SPI1_NSS    (HV2 DAC CS)
PA05  SPI1_SCK    (HV2 DAC SCLK)
PA07  SPI1_MOSI   (HV2 DAC MOSI)
PA13  SWDIO       (debug)
PA14  SWCLK       (debug)

PB02  B00T1       (boot strap)
PB08  I2C1_SCL    (SHT31)
PB09  I2C1_SDA    (SHT31)
PB12  SPI2_NSS    (HV1 DAC CS)
PB13  SPI2_SCK    (HV1 DAC SCLK)
PB15  SPI2_MOSI   (HV1 DAC MOSI)

PC04  HV2_R/S     (HV2 run/stop)
PC10  USART3_TX   (console)
PC11  USART3_RX   (console)

PD09  HV1_R/S     (HV1 run/stop)

PE00  I2C1_ALERT  (SHT31 alert)
PE07  HV1_ADC_A1
PE08  HV1_ADC_A0
PE10  HV1_ADC_GAIN1
PE11  HV1_ADC_CS  (/PDWN)
PE12  HV1_ADC_SCLK
PE13  HV1_ADC_MISO (DRDY)

```

PF03 HV2_ADC_A0
 PF04 HV2_ADC_GAIN1
 PF06 HV2_ADC_CS (/PDWN)
 PF07 HV2_ADC_SCLK
 PF08 HV2_ADC_MISO (DRDY)
 PF10 HV2_ADC_GAIN0

 PG09 USART6_RX (RS-485)
 PG11 RS485_DIR (RS-485 DE/RE)
 PG14 USART6_TX (RS-485)

 PH12 HV1_ADC_GAIN0

 PI05 SYS_MOD0 (DIP switch)
 PI07 SYS_MOD1 (DIP switch)
 PI12 SYS_RUN (status LED)
 PI14 HV2_ADC_A1

 B00T0 boot strap
 NRST reset

Total: 36 MCU pins consumed (excluding VDD/VSS/VCAP).

1.4 4. Available Pins

GPIOA–GPIOI are enabled in board DTS (status = “okay”). GPIOJ/GPIOK exist in the SoC but are **not enabled** in the board DTS — they may not be broken out on the PCB. Pin counts below exclude used pins listed in Section 3.

Port	Free pins	Count
GPIOA	PA0,PA1,PA2,PA3,PA6,PA8,PA9,PA10,PA11,PA12,PA15	11
GPIOB	PB0,PB1,PB3,PB4,PB5,PB6,PB7,PB10,PB11,PB14	10
GPIOC	PC0,PC1,PC2,PC3,PC5,PC6,PC7,PC8,PC9,PC12,PC13,PC14,PC15	13
GPIOD	PD0,PD1,PD2,PD3,PD4,PD5,PD6,PD7,PD8,PD10,PD11,PD12,PD13,PD14,PD15	15
GPIOE	PE1,PE2,PE3,PE4,PE5,PE6,PE9,PE14,PE15	9
GPIOF	PF0,PF1,PF2,PF5,PF9,PF11,PF12,PF13,PF14,PF15	10
GPIOG	PG0,PG1,PG2,PG3,PG4,PG5,PG6,PG7,PG8,PG10,PG12,PG13,PG15	13
GPIOH	PH0,PH1,PH2,PH3,PH4,PH5,PH6,PH7,PH8,PH9,PH10,PH11,PH13,PH14,PH15	15
GPIOI	PI0,PI1,PI2,PI3,PI4,PI6,PI8,PI9,PI10,PI11	10
Total		~104

Note: Not all STM32F429BI LQFP208 pins may be physically routed to headers on the jw_hvb PCB. Verify with board layout before finalizing expansion connector pinout.

1.5 5. Solutions Comparison

1.5.1 5.1 BME280 — Interface Options

1.5.1.1 Option A: Share I2C1 (0 extra pins) ★ Recommended

Net	MCU pin	Bus	Shared with
I2C1_SCL	PB8	I2C1	SHT31 (0x44)
I2C1_SDA	PB9	I2C1	SHT31 (0x44)

- **Pros:** Zero additional pins. Already-enabled bus. No DT change to I2C1 controller node. BME280 addr 0x76/0x77 does not collide with SHT31 at 0x44. Zephyr I2C API serializes transactions per bus — no race between sensor reads. Kconfig: just add CONFIG_BME280=y.
- **Cons:** Both sensors share I2C1 bandwidth. SHT31 single-shot takes ~15 ms; BME280 forced mode takes ~8 ms. Combined worst-case poll latency ~23 ms — negligible for environmental sensing at 1 Hz. Bus fault from one device can stall the other (I2C has no reset mechanism on STM32F4 v1 peripheral).
- **Verdict:** Best choice. Adds zero pins. Bandwidth is not a concern for <10 Hz sensor polling.

1.5.1.2 Option B: Dedicated I2C2 (2 extra pins)

Net	MCU pin	Bus
I2C2_SCL	PB10	I2C2 (AF4)
I2C2_SDA	PB11	I2C2 (AF4)

- **Pros:** Electrical isolation from SHT31. No shared bandwidth. Independent bus recovery.
- **Cons:** 2 extra pins consumed. PB10/PB11 would displace rotary encoder SW and BTN1 from the recommended PB cluster. I2C2 must be enabled in DT. Zephyr stm32-i2c-v1 handles multiple instances fine.
- **Verdict:** Overkill for two low-data-rate sensors. Only justified if galvanic separation or redundant sensor paths are required.

1.5.1.3 Option C: SPI (4 extra pins)

Net	MCU pin	Bus
BME_CS	any GPIO	—
SPIx_SCK	SPI pin	SPIx
SPIx_MOSI	SPI pin	SPIx
SPIx_MISO	SPI pin	SPIx

- **Pros:** Higher throughput (10 MHz vs 400 kHz I2C). No address collision risk.
- **Cons:** 4 extra pins. Must share or dedicate an SPI bus (all 3 free SPIs have pin conflicts — see §5.2). BME280 throughput needs are trivial (~24 bytes per read). Overcomplicates the design.
- **Verdict:** Not justified. I2C is the natural interface for BME280 in this design.

Criterion	A: Shared I2C1	B: Dedicated I2C2	C: SPI
Pins	0	2	4
Firmware complexity	Low	Low	Medium
Bus isolation	Shared with SHT31	Isolated	Isolated
Bandwidth risk	Negligible	None	None
PCB trace impact	None (existing)	2 new traces	4 new traces

1.5.2 5.2 SSD1331 OLED — Interface Options

1.5.2.1 Option A: SPI3 on PB3/PB5 (5 pins) ★ Recommended

Signal	MCU pin	AF
SCLK	PB3	SPI3_SCK (AF6)
MOSI	PB5	SPI3_MOSI (AF6)
CS	PB0	GPIO
DC	PB1	GPIO
RST	PB4	GPIO

- **Pros:** Dedicated SPI bus — no timing sharing with DACs or ADCs. All 5 pins cluster on GPIOB (PB0–PB5) for clean routing. SPI3 is fully free (not used by any existing device). Follows cs-gpios pattern already used for AD5541 DACs on SPI1/SPI2. 45 Mbit/s capable (STM32F4 SPI), far exceeding SSD1331 max (~10 MHz).
- **Cons:** Consumes 5 pins. SPI3 consumes an APB1 clock gate (minor power).
- **Verdict:** SPI3 is the only SPI peripheral with both SCK and MOSI pins fully free. Best choice.

1.5.2.2 Option B: Bit-bang GPIO (5 pins, any port)

Signal	MCU pin	Mode
SCLK	Any free GPIO	GPIO output
MOSI	Any free GPIO	GPIO output
CS	Any free GPIO	GPIO output
DC	Any free GPIO	GPIO output
RST	Any free GPIO	GPIO output

- **Pros:** Total pin placement freedom — pick any 5 free GPIOs on any port. Follows the ADS1232 bit-bang pattern already established in this design. No hardware SPI peripheral consumed.
- **Cons:** Lower throughput — GPIO bit-bang peaks at ~2–4 MHz vs SPI hardware at 45 Mbit/s. SSD1331 needs $\sim 96 \times 64 \times 16$ bits = 12.3 KB per full-screen refresh. At 2 MHz GPIO, that's ~50 ms per refresh — still acceptable for a UI panel. CPU spin-waits during transfer (no DMA); for a 50 ms transfer once per second this is negligible.
- **Verdict:** Viable fallback if SPI3 pins are unavailable on PCB. Performance difference is not user-visible for a status display.

1.5.2.3 Option C: Share SPI1 or SPI2 with existing DAC (3 extra pins)

Net	MCU pin	Shared bus
OLED_CS	GPIO	—
OLED_DC	GPIO	—
OLED_RST	GPIO	—
SCLK	PA5 or PB13	SPI1 (HV2 DAC) or SPI2 (HV1 DAC)
MOSI	PA7 or PB15	SPI1 (HV2 DAC) or SPI2 (HV1 DAC)

- **Pros:** Saves 2 pins (SCLK, MOSI reused). No new SPI peripheral needed.

- **Cons:** HV DAC timing is safety-critical — an OLED frame write (~12 KB) could stall a DAC update for ~50 ms. The 10 Hz LPF after DAC smooths this, but it introduces coupling between UI and HV control paths. Requires cs-gpios array management. Architecturally unclear: mixing display and safety-critical HV control on same SPI bus.
- **Verdict:** Not recommended. Safety-critical HV DAC timing should not share a bus with a display.

1.5.2.4 Option D: SPI6 (5 pins)

Signal	MCU pin	AF
SCLK	PG13	SPI6_SCK (AF5)
MOSI	PG14	SPI6_MOSI (AF5) — conflict

- **Cons:** PG14 is already USART6_TX (RS-485 Modbus). Cannot be remapped without breaking Modbus. Non-starter.
- **Verdict:** Rejected.

Criterion	A: SPI3 PB3/PB5	B: Bit-bang GPIO	C: Share SPI1/2	D: SPI6
Pins	5	5	3	5
Max throughput	45 Mbit/s	~2-4 Mbit/s	45 Mbit/s	45 Mbit/s
Bus contention	None	None	Shares with DAC	None
Safety isolation	Clean	Clean	Risky	N/A
Firmware complexity	Medium (SPI3 init)	Low (GPIO toggle)	Medium (CS sharing)	—
Pin cluster	PB0-PB5 (same port)	Any port	Scattered	Scattered
DMA capable	Yes	No	Yes	—
Conflict with existing	None	None	None if SPI2 CS timing OK	PG14 = USART6

1.5.3 5.3 Rotary Encoder — Interface Options

1.5.3.1 Option A: 2 GPIO + Interrupt (3 pins) ★ Recommended

Signal	MCU pin	Mode
ENC_A	PB6	GPIO input, interrupt on both edges
ENC_B	PB7	GPIO input
ENC_SW	PB10	GPIO input, interrupt on falling edge

- **Pros:** Simple. Any 3 free GPIOs work. Software state machine decodes quadrature — typical implementation is ~30 lines of C. Interrupt on ENC_A edge reads ENC_B level to determine direction. No hardware timer consumed.
- **Cons:** ISR overhead per encoder tick. At 24 PPR detents × 4 edges = 96 interrupts per revolution. Human knob turning max ~5 rev/s → ~480 interrupts/s — negligible on 168 MHz Cortex-M4F. Debounce needed (hardware RC or software timer).
- **Verdict:** Good enough. Interrupt rate is trivial. Software debounce is simple.

1.5.3.2 Option B: TIM4 QDEC Mode (2 pins, hardware decode)

Signal	MCU pin	AF
ENC_A	PB6	TIM4_CH1 (AF2)
ENC_B	PB7	TIM4_CH2 (AF2)
ENC_SW	PB10	GPIO input

- **Pros:** Hardware quadrature decode — no ISR per count edge. TIM4 counter directly holds position. Zephyr has `st,stm32-qdec` binding on TIM4. Filter/debounce in hardware (`st,input-filter-level`). Zero CPU overhead for counting.
- **Cons:** Consumes TIM4 peripheral (TIM4 is free on `jw_hvb`). PB6/PB7 must be mapped to AF2, tying these specific pins. If PB6/PB7 are unavailable on PCB, must use PD12/PD13 as TIM4 CH1/CH2 alternative. Zephyr QDEC driver must be enabled (`CONFIG_QDEC=y`, `CONFIG_QDEC_STM32=y`).
- **Verdict:** Preferred if PB6/PB7 are confirmed routed. Lower power (no ISR wake), cleaner code. GPIO fallback if QDEC driver has issues.

Criterion	A: GPIO + ISR	B: TIM4 QDEC
Pins	3	3
CPU overhead	~480 ISR/s	0
Hardware cost	None	Consumes TIM4
Debounce	Software	Hardware filter
Pin flexibility	Any GPIOs	Must be PB6/PB7 or PD12/PD13
Firmware complexity	Medium (state machine)	Low (counter read)
Zephyr support	Manual	<code>st,stm32-qdec</code> driver

1.5.4 5.4 Four Buttons — Interface Options

1.5.4.1 Option A: 4 Individual GPIOs (4 pins) ★ Recommended

Button	MCU pin
BTN1	PB11
BTN2	PB14
BTN3	PC0
BTN4	PC1

- **Pros:** Simple. Each button independently interrupt-capable (wake from sleep). No analog noise concerns. Zephyr `gpio-keys` binding directly supports this pattern. Independent debounce per pin.
- **Cons:** 4 pins consumed.
- **Verdict:** With ~104 free pins, 4 GPIOs is not a scarce resource. Best choice.

1.5.4.2 Option B: ADC Resistor Ladder (1 pin)

Signal	MCU pin
BTN_ADC	Any ADC input (e.g., PA0/ADC123_IN0)

- **Pros:** Only 1 pin. Classic low-pin-count technique. STM32F4 ADC is 12-bit — enough to distinguish 4 buttons + idle.
- **Cons:** Requires calibration. Analog noise couples into button detection. Cannot detect simultaneous presses reliably. Each button press draws current through divider. ADC must be enabled (CONFIG_ADC=y). Slower polling vs interrupt-driven GPIO. More complex firmware (voltage threshold table).
- **Verdict:** Saving 3 pins is not worth the analog complexity on this board. Only justified if pin count is truly critical.

1.5.4.3 Option C: I2C GPIO Expander (0 extra MCU pins)

Net	MCU pin
I2C1_SCL	PB8 (shared)
I2C1_SDA	PB9 (shared)

- **Pros:** Zero MCU pins. Supports up to 8+ buttons with interrupt output. Clean.
- **Cons:** Adds BOM cost (\$0.30–0.80 for PCA9536/TCA9534). Adds I2C bus traffic. Adds firmware driver dependency. Power draw (~1 μ A standby). Over-engineered for 4 buttons.
- **Verdict:** Not warranted. 4 GPIOs are cheaper and simpler than an expander chip.

Criterion	A: 4 GPIOs	B: ADC ladder	C: I2C expander
Pins	4	1	0
BOM cost	4 $\times$ pull-up R	5 $\times$ R + calibration	+\$0.30–0.80 IC
Simultaneous press	Yes	No	Yes
Interrupt wake	Yes (per pin)	No (poll ADC)	Yes (INT pin)
Noise immunity	High	Low (analog)	High
Firmware complexity	Low	Medium	Medium
PCB area	4 traces	1 trace + 4 R	IC + 2 traces

1.5.5 5.5 Overall Architecture — Top-Level Comparison

1.5.5.1 Solution 1: Balanced (Recommended) — 12 pins

Device	Interface	Pins
BME280	I2C1 shared	0
SSD1331 OLED	SPI3 HW	5
Rotary encoder	GPIO ISR	3
4 buttons	4 $\times$ GPIO	4
Total		12

- Clean separation: display on SPI3, sensors on I2C1, HMI on GPIO.
- All pins cluster on GPIOB + GPIOC for easy routing.
- Follows established patterns (DAC=SPI, ADC=GPIO bit-bang, sensor=I2C).

1.5.5.2 Solution 2: Pin-Minimal — 8 pins

Device	Interface	Pins
BME280	I2C1 shared	0
SSD1331 OLED	Bit-bang GPIO	5
Rotary encoder	GPIO ISR	3
4 buttons	ADC ladder	1
Total		9

- Saves 3 pins vs Solution 1 by using ADC ladder for buttons.
- Sacrifices: slower display refresh (bit-bang), analog noise on buttons, no simultaneous key detection.
- Only justified if expansion connector is size-constrained (e.g., 10-pin FFC instead of 16-pin header).

1.5.5.3 Solution 3: Pin-Rich (Clean Isolation) — 14 pins

Device	Interface	Pins
BME280	I2C2 dedicated	2
SSD1331 OLED	SPI3 HW	5
Rotary encoder	TIM4 QDEC	3
4 buttons	4× GPIO	4
Total		14

- Maximum isolation: every device on its own bus.
- Costs 2 extra pins vs Solution 1 for I2C2.
- Only justified if BME280/SHT31 bus sharing proves problematic in testing.

Criterion	Sol 1: Balanced	Sol 2: Pin-minimal	Sol 3: Pin-rich
Total pins	12	9	14
Display speed	HW SPI (fast)	Bit-bang (slow)	HW SPI (fast)
Encoder decode	SW ISR	SW ISR	HW QDEC (zero CPU)
Button detection	Individual	ADC (analog)	Individual
Sensor isolation	Shared I2C1	Shared I2C1	Dedicated I2C2
Pin clustering	Good (PB+PC)	Any port	Good
Recommended for	Default	Space-constrained	Max reliability

1.6 6. Recommended Configuration (Solution 1)

Pin selections are optimized for: - Physical adjacency (same port cluster → easier PCB routing on expansion board) - Hardware peripheral availability (SPI3 for display, optional TIM4_QDEC for encoder) - No conflict with existing or planned usage

1.6.1 6.1 BME280 — Share I2C1 (0 extra pins)

BME280 at 0x76 or 0x77 coexists with SHT31 at 0x44 on the same I2C1 bus.

Net	MCU pin	Already used by
I2C1_SCL	PB8	SHT31
I2C1_SDA	PB9	SHT31

No firmware/DT change to the I2C1 bus — add a second child node under &i2c1.

1.6.2 6.2 SSD1331 OLED — SPI3 (5 pins)

Signal	MCU pin	AF / mode	Notes
SCLK	PB3	AF6 (SPI3_SCK)	Hardware SPI clock
MOSI (DIN)	PB5	AF6 (SPI3_MOSI)	Hardware SPI data
CS	PB0	GPIO output (active low)	Software chip select
DC	PB1	GPIO output	Data/Command select
RST	PB4	GPIO output (active low)	Hardware reset

All 5 pins are on GPIOB, adjacent (PB0–PB5). SPI3 is unused in the current design. The SSD1331 is write-only (no MISO needed), matching the AD5541 DAC pattern already used on SPI1/SPI2.

1.6.3 6.3 Rotary Encoder (3 pins)

Signal	MCU pin	Type	Notes
ENC_A	PB6	GPIO input, interrupt	Also TIM4_CH1 (AF2) for QDEC
ENC_B	PB7	GPIO input, interrupt	Also TIM4_CH2 (AF2) for QDEC
ENC_SW	PB10	GPIO input, interrupt	Push-button

PB6/PB7 optionally map to TIM4 QDEC for hardware quadrature decoding (preferred if QDEC driver works; fall back to GPIO ISR).

1.6.4 6.4 Four Buttons (4 pins)

Button	MCU pin	Type
BTN1	PB11	GPIO input, pull-up
BTN2	PB14	GPIO input, pull-up
BTN3	PC0	GPIO input, pull-up
BTN4	PC1	GPIO input, pull-up

All buttons: active low with external or internal pull-up (internal pull-ups on STM32F4 are ~40 kΩ; add external 10 kΩ if noise is a concern).

1.6.5 6.5 Pin Summary

Port B cluster:

PB0 → OLED_CS
PB1 → OLED_DC
PB3 → OLED_SCLK (SPI3_SCK)
PB4 → OLED_RST
PB5 → OLED_MOSI (SPI3_MOSI)
PB6 → ENC_A
PB7 → ENC_B
PB8 → I2C1_SCL (shared: SHT31 + BME280)
PB9 → I2C1_SDA (shared: SHT31 + BME280)
PB10 → ENC_SW
PB11 → BTN1
PB14 → BTN2

Port C cluster:

PC0 → BTN3
PC1 → BTN4

Device	New MCU pins
BME280	0 (shares I2C1)
SSD1331 OLED	5
Rotary encoder	3
4 buttons	4
Total	12

1.7 7. Hardware Design Details

1.7.1 7.1 BME280 Environmental Sensor

1.7.1.1 Module Selection

Module	Dimensions	Notes
GY-BME280	15 × 11.5 mm	Common breakout, 3.3V/5V via onboard LDO, I2C+SPI
CJMCU-280	13 × 10 mm	Compact, 3.3V only, I2C only
Custom PCB	—	Bare BME280 IC (LGA-8 2.5×2.5×0.93 mm)

Recommendation: **GY-BME280** module (widely available, well-documented).

1.7.1.2 Pinout (GY-BME280)

Module pin	Connect to	Notes
VCC	VCC_+3V3	Module has onboard 3.3V LDO; can also bypass LDO
GND	GND	
SCL	I2C1_SCL (PB8)	Shared with SHT31

Module pin	Connect to	Notes
SDA	I2C1_SDA (PB9)	Shared with SHT31
CSB	VCC_+3V3	Tie HIGH to select I2C mode
SDO	GND or VCC_+3V3	GND → addr 0x76; VCC → addr 0x77

1.7.1.3 I2C Address Selection Place a solder jumper on the expansion PCB to select SDO level:

SDO — 2-pin header (jumper open → pull-up to VCC = 0x77)
 └─ 2-pin header (jumper closed → pull-down to GND = 0x76, default)

Default: **0x76** (SDO = GND). Ship with jumper closed or a 10 kΩ pull-down to GND.

1.7.1.4 I2C Bus Pull-up Check Existing SHT31 on jw_hvb uses R96/R97 = **10 kΩ** pull-ups on SCL/SDA (see ref/board_design.md §9). Adding BME280 on the same bus:

- STM32F4 I2C1 fast-mode (400 kHz) requires $R_p(\min) \approx 1.3 \text{ k}\Omega$, $R_p(\max) \approx 3.6 \text{ k}\Omega$ at 3.3V
- Current 10 kΩ is too weak for 400 kHz — the bus is currently running at **standard-mode (100 kHz)** per DT (clock-frequency = <I2C_BITRATE_STANDARD>)
- At 100 kHz with 10 kΩ: rise time $\approx 10\text{k} \times 30\text{pF} \times 3 \approx 0.9 \mu\text{s}$ — marginal but works for SHT31
- Adding BME280 adds $\sim 5 \text{ pF}$ bus capacitance → still OK at 100 kHz
- **No change needed.** If you want 400 kHz in future, replace R96/R97 with 2.2 kΩ.

1.7.1.5 Decoupling

Component	Value	Package	Placement
C_BME280	100 nF X7R	0402 or 0603	≤5 mm from module VCC pin

1.7.2 7.2 SSD1331 OLED Display

1.7.2.1 Module Selection

Module	P/N	Dimensions	Resolution	Colors
Waveshare 0.95" OLED	SKU 14731	31.5 × 27.0 mm	96×64	65K/262K
Generic SSD1331 OLED	—	~26 × 20 mm	96×64	65K

Recommendation: Waveshare 0.95inch RGB OLED (HAT) — well-documented pinout, 3.3V compatible, mounting holes provided.

1.7.2.2 Module Pinout (Waveshare)

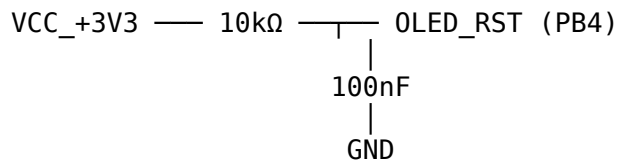
Pin	Name	Connect to	Notes
1	VCC	VCC_+3V3	Module accepts 3.3V directly
2	GND	GND	

Pin	Name	Connect to	Notes
3	DIN	SPI3_MOSI (PB5)	Data input (MOSI)
4	CLK	SPI3_SCLK (PB3)	SPI clock
5	CS	OLED_CS (PB0)	Active-low chip select
6	DC	OLED_DC (PB1)	Data/Command: LOW=cmd, HIGH=data
7	RST	OLED_RST (PB4)	Active-low reset

1.7.2.3 Electrical Characteristics

Parameter	Value	Notes
VCC supply	3.3 V $\pm$ 5%	Module has onboard regulator
I_VCC (typical)	25 mA	All pixels white, max brightness
I_VCC (sleep)	<10 μ A	Display off via SW command
V_IH (logic HIGH)	$\geq 0.7 \times VCC$	2.31 V min
V_IL (logic LOW)	$\leq 0.3 \times VCC$	0.99 V max
SPI max clock	10 MHz	Mode 0 (CPOL=0, CPHA=0)
Reset low pulse	$\geq 3 \mu$ s	After VCC stable ≥ 100 ms

1.7.2.4 Hardware Reset Circuit Add an RC delay on the expansion PCB for reliable power-on reset:



This holds RST low for ~ 1 ms after power-up. PB4 GPIO can also drive RST for software-triggered reset through a series 1k Ω resistor.

1.7.2.5 Decoupling

Component	Value	Package	Placement
C_OLED1	100 nF X7R	0402	≤ 5 mm from module VCC pin
C_OLED2	10 μ F X5R	0805	≤ 10 mm from module VCC pin

1.7.3 7.3 Rotary Encoder

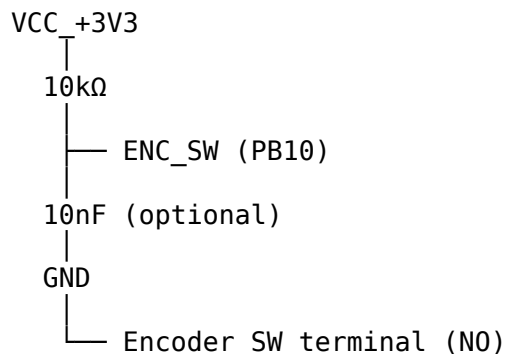
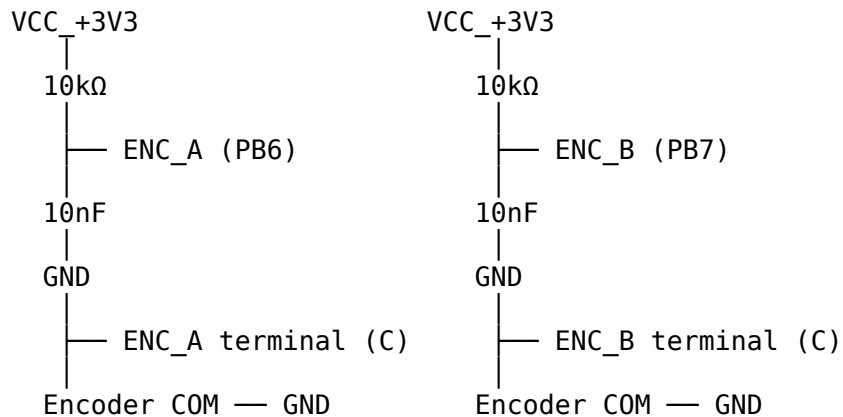
1.7.3.1 Part Selection

Part	Manufacturer	PPR	Detents	Shaft	Notes
EC11E15244A2	ALPS	15 PPR	30	20 mm flatted	Good feel, common
PEC11R-4220F-S0024	Bourns	24 PPR	24	20 mm flatted	Higher resolution

Part	Manufacturer	PPR	Detents	Shaft	Notes
KY-040 module	Generic	20 PPR	20	module	Inexpensive breakout with RC circuit

Quadrature output: Channel A leads Channel B by 90° for clockwise rotation.

1.7.3.2 Reference Circuit



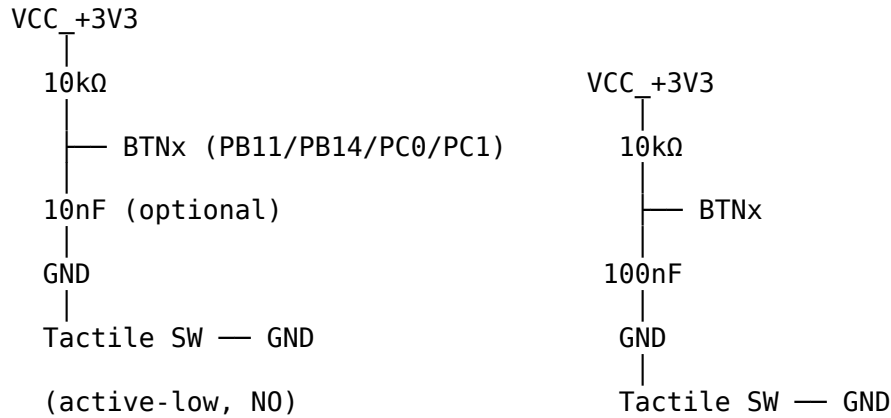
- **Pull-up:** 10 kΩ external. Can also use STM32 internal pull-up (~40 kΩ) — skip external R if using internal.
- **Debounce RC:** $10\text{ k}\Omega \times 10\text{ nF} = 100\text{ }\mu\text{s}$ time constant. For 24 PPR at 5 rev/s, max pulse rate = 120 Hz, period = 8.3 ms. 100 μs debounce is 1.2% of pulse width — safe. If using TIM4 QDEC, enable hardware filter (`st, input-filter-level = <5>` for ~500 ns filtering).
- **Alternative:** If using internal pull-ups, place only 10 nF caps (no external R).

1.7.3.3 Mechanical

- Panel-mount encoder with nut. Shaft diameter typically 6 mm.
- Standard knob cap fits 6 mm D-shaft.
- Module placement: edge of expansion PCB with shaft protruding through enclosure.

1.7.4 7.4 Tactile Buttons

1.7.4.1 Reference Circuit



- **Pull-up:** 10 kΩ external recommended (STM32 internal ~40 kΩ is OK for bench testing but vulnerable to EMI in deployed environment).
- **Debounce:** 100 nF across switch (10 kΩ × 100 nF = 1 ms time constant) provides clean hardware debounce. Software debounce still recommended as backup.
- **ESD protection:** If buttons are user-facing and exposed to touch, add a TVS array:
 - USBLC6-2SC6 (SOT-23-6, 2 channels) per 2 buttons
 - Or discrete: BAT54S dual Schottky + 100 nF on each button line

1.7.4.2 Recommended Switch

Part	Manufacturer	Size	Actuation force	Travel
PTS645SM43SMTR92	C&K	6×6 mm	160 gf	0.25 mm
B3F-1000	Omron	6×6 mm	100 gf	0.25 mm
KSC421J	C&K	6.2×6.2 mm	200 gf	0.35 mm

Any SPST-NO tactile switch with 6×6 mm footprint, 4–5 mm height.

1.7.5 7.5 Pin Electrical Characteristics (STM32F429BIT6 Side)

All recommended pins below are 5 V-tolerant except where noted.

MCU pin	Signal	5V-tolerant	Max output speed	PU/PD	Notes
PB0	OLED_CS	Yes	100 MHz	PU/PD	GPIO output GPIO output SPI3_SCK, also JTDO (disabled in SWD mode)
PB1	OLED_DC	Yes	100 MHz	PU/PD	
PB3	OLED_SCLK	Yes	100 MHz	PU	
PB4	OLED_RST	Yes	100 MHz	PU	GPIO output, also NJTRST (disabled in SWD mode)

MCU pin	Signal	5V-tolerant	Max output speed	PU/PD	Notes
PB5	OLED_MOSI	Yes	100 MHz	PU	SPI3_MOSI
PB6	ENC_A	Yes	100 MHz	PU/PD	GPIO input, also TIM4_CH1
PB7	ENC_B	Yes	100 MHz	PU/PD	GPIO input, also TIM4_CH2
PB8	I2C1_SCL	Yes	100 MHz	PU/PD	I2C1 (FT — 5V tolerant), open-drain required
PB9	I2C1_SDA	Yes	100 MHz	PU/PD	I2C1 (FT — 5V tolerant), open-drain required
PB10	ENC_SW	Yes	100 MHz	PU/PD	GPIO input
PB11	BTN1	Yes	100 MHz	PU/PD	GPIO input
PB14	BTN2	Yes	100 MHz	PU/PD	GPIO input
PC0	BTN3	Yes	100 MHz	PU/PD	GPIO input
PC1	BTN4	Yes	100 MHz	PU/PD	GPIO input

Note: PB3 and PB4 are JTAG pins (JTDO/NJTRST). In SWD mode (used by jw_hvb via PA13/PA14 SWDIO/SWCLK), these pins are available as GPIO. Ensure the debug probe does not assert JTAG mode. This is the default for STM32F4 when SWD is used.

1.7.6 7.6 Signal Integrity & PCB Layout Guidelines

1.7.6.1 I2C Bus (PB8, PB9)

- Trace impedance: not critical at 100 kHz
- Keep SCL and SDA traces together, length-matched within 20 mm
- Total bus capacitance: keep <400 pF (SHT31 $\approx$ 5 pF, BME280 $\approx$ 5 pF, traces $\approx$ 2 pF/cm)
- Pull-ups (R96/R97) are on the jw_hvb main board (10 k Ω). Do NOT add duplicate pull-ups on the expansion PCB.

1.7.6.2 SPI Bus (PB3, PB5)

- Target: 8–10 MHz operation
- Trace impedance: 50 Ω nominal (not critical at these speeds)
- Keep SCK and MOSI trace lengths matched within 10 mm
- Maximum trace length: 100 mm
- For traces >50 mm: add 22 Ω series termination resistor near MCU-side connector on SCK line to damp reflections
- No MISO on this bus (SSD1331 is write-only)

1.7.6.3 GPIO Lines (PB0, PB1, PB4, PB6, PB7, PB10, PB11, PB14, PC0, PC1)

- No special routing requirements
- Keep >2 mm from any switching HV traces (if expansion PCB is near HV section)
- Buttons: use ground pour around switch pads for ESD protection

1.7.6.4 Power Distribution

- VCC_+3V3 trace width: ≥ 0.5 mm for 100 mA capacity
- GND: use ground plane on bottom layer of expansion PCB
- Place a 10 μ F bulk capacitor at the connector entry point
- Each device gets a local 100 nF decoupling capacitor

1.7.6.5 General

- 2-layer PCB sufficient for all signals
- Expansion PCB ground plane must connect to jw_hvb GND at the connector (at least 2 GND pins)
- If expansion PCB is >50 mm from jw_hvb, consider adding a ferrite bead (e.g., BLM18PG121SN1) on VCC_+3V3 at the expansion PCB entry

1.7.7 7.7 Expansion PCB Reference Schematic Checklist

- ☐ Connector: 16-pin 2x8 header (see §8)
- ☐ Jumper: BME280 SDO address select (0x76 default)
- ☐ BME280: 100 nF decoupling cap at module VCC
- ☐ SSD1331: 100 nF + 10 μ F decoupling at module VCC
- ☐ SSD1331: RC reset circuit (10k Ω + 100nF) on RST line
- ☐ Encoder: 10k Ω pull-ups on ENC_A, ENC_B, ENC_SW
- ☐ Encoder: 10 nF debounce caps (optional, if not using internal pull-ups + SW debounce)
- ☐ Buttons: 10k Ω pull-ups + 100 nF debounce caps on each BTNx
- ☐ Buttons: TVS ESD protection if user-exposed (e.g., USBLC6-2SC6)
- ☐ Power: 10 μ F bulk cap at connector VCC entry
- ☐ Test points: GND test point (for scope probe) near connector

1.7.8 7.8 Bill of Materials (Expansion PCB)

Ref	Qty	Value / Part	Package	Notes
U1	1	GY-BME280 module	15x11.5 mm	BME280 environmental sensor breakout
U2	1	Waveshare SSD1331 0.95" OLED	31.5x27 mm	Or generic SSD1331 module
SW_ENC	1	EC11E15244A2 or PEC11R-4220F	Panel mount	Rotary encoder with push switch
SW1-SW4	4	PTS645SM43SMTR	5x6 mm SMD	Tactile switch, SPST NO
J1	1	2x8 pin header, 2.54 mm	TH/SMD	To jw_hvb (see §8 for mating)
R1-R3	3	10 k Ω $\pm 5\%$	0402 or 0603	Encoder pull-ups (omit if using MCU internal)
R4-R7	4	10 k Ω $\pm 5\%$	0402 or 0603	Button pull-ups

Ref	Qty	Value / Part	Package	Notes
R8	1	10 k Ω $\pm$ 5%	0402 or 0603	OLED RST RC circuit
R9	1	1 k Ω $\pm$ 5%	0402 or 0603	OLED RST series resistor (SW reset)
R10	1	10 k Ω $\pm$ 5%	0402 or 0603	BME280 SDO pull-down (addr 0x76)
C1	1	100 nF X7R	0402	BME280 decoupling
C2	1	100 nF X7R	0402	OLED decoupling
C3	1	10 μ F X5R	0805	OLED bulk decoupling
C4	1	10 μ F X5R	0805	Connector VCC entry bulk decoupling
C5	1	100 nF X7R	0402	OLED RST RC delay cap
C6-C9	4	100 nF X7R	0402	Button debounce caps (optional)
C10-C11	2	10 nF X7R	0402	Encoder debounce caps (optional)
D1-D2	2	USBL C6-2SC6	SOT-23-6	ESD protection for 4 buttons (optional)
JP1	1	2-pin header, 2.54 mm	TH	BME280 SDO address jumper

1.8 8. Expansion Connector Specification

1.8.1 8.1 Recommended Connectors

Option	Series	Pitch	Pins	Mating height	Notes
A	Standard pin header	2.54 mm	2 $\times$ 8	8.5 mm	Simplest, breadboard-friendly
B	JST SH (SM16B-SRSS-TB)	1.00 mm	16 (single row)	4.5 mm	Compact, locking, good for production
C	Molex PicoBlade 53398	1.25 mm	16	4.2 mm	Compact, locking
D	Harwin M20	2.54 mm	2 $\times$ 8	8.5 mm	Industrial, shrouded, polarized

Recommendation for development: Option A (standard 2.54 mm pin header). Cheap, easy to probe. **Recommendation for production:** Option B (JST SH 1.0 mm). Compact, locking, vibration-resistant.

1.8.2 8.2 Connector Pinout

Pin	Signal	MCU pin	Direction	Option A pin	Option B pin
1	VCC_+3V3	—	Power out	Row A, Col 1	Pin 1
2	GND	—	—	Row A, Col 2	Pin 2
3	I2C1_SCL	PB8	Bidir (OD)	Row A, Col 3	Pin 3
4	I2C1_SDA	PB9	Bidir (OD)	Row A, Col 4	Pin 4
5	SPI3_SCLK	PB3	Output	Row A, Col 5	Pin 5
6	SPI3_MOSI	PB5	Output	Row A, Col 6	Pin 6
7	OLED_CS	PB0	Output	Row A, Col 7	Pin 7
8	OLED_DC	PB1	Output	Row A, Col 8	Pin 8
9	OLED_RST	PB4	Output	Row B, Col 1	Pin 9
10	ENC_A	PB6	Input	Row B, Col 2	Pin 10
11	ENC_B	PB7	Input	Row B, Col 3	Pin 11
12	ENC_SW	PB10	Input	Row B, Col 4	Pin 12
13	BTN1	PB11	Input	Row B, Col 5	Pin 13
14	BTN2	PB14	Input	Row B, Col 6	Pin 14
15	BTN3	PC0	Input	Row B, Col 7	Pin 15
16	BTN4	PC1	Input	Row B, Col 8	Pin 16

Mating connectors (for expansion PCB side):

Option	Connector	Part number
A	2×8 pin header female	e.g., Samtec SSW-108-01-G-D (TH) or equivalent
A	16-pin IDC ribbon	2×8 IDC socket + ribbon cable
B	JST SH receptacle	SHR-16V-S-B, plus SHF-001T-0.8BS crimp contacts
C	Molex PicoBlade receptacle	51021-1600, plus 50079-8100 crimp contacts

1.9 9. Fallback Pin Options

Original	Fallback	Reason
PB3 (SPI3_SCK)	PC10 (SPI3_SCK, AF6)	Conflicts with USART3_TX unless moved
PB5 (SPI3_MOSI)	PC12 (SPI3_MOSI, AF6)	—
PB0 (OLED_CS)	Any free GPIO	Software CS is portable
PB1 (OLED_DC)	Any free GPIO	—
PB4 (OLED_RST)	Any free GPIO	—

Original	Fallback	Reason
PB6 (ENC_A)	PD12 (TIM4_CH1, AF2)	If PB6 is unavailable
PB7 (ENC_B)	PD13 (TIM4_CH2, AF2)	If PB7 is unavailable

SPI4/SPI5 alternative (if SPI3 pins are blocked): - SPI4: PE12 (SCK), PE14 (MISO), PE6 (MOSI) — but PE12 is used by HV1 ADC - SPI5: PF7 (SCK), PF8 (MISO), PF9 (MOSI) — but PF7/PF8 are used by HV2 ADC - SPI6: PG12 (MISO), PG13 (SCK), PG14 (MOSI) — PG14 is USART6_TX

If all hardware SPI buses are blocked: Bit-bang SSD1331 on any 5 free GPIOs (follows the ADS1232 pattern already established in the design).

1.10 10. Firmware Impact

1.10.1 10.1 Devicetree Changes

```
// 1. BME280 on existing I2C1 bus
&i2c1 {
    bme280@76 {
        compatible = "bosch,bme280";
        reg = <0x76>;
    };
};

// 2. Enable SPI3
&spi3 {
    pinctrl-0 = <&spi3_sck_pb3 &spi3_mosi_pb5>;
    pinctrl-names = "default";
    cs-gpios = <&gpio0 0 GPIO_ACTIVE_LOW>;
    status = "okay";
    // SSD1331 display (write-only, no kernel driver in Zephyr –
    // use display API or custom driver)
};

&pinctrl {
    spi3_sck_pb3: spi3_sck_pb3 { pinmux = <STM32_PINMUX('B', 3, AF6)>; };
    spi3_mosi_pb5: spi3_mosi_pb5 { pinmux = <STM32_PINMUX('B', 5, AF6)>; };
};

// 3. Rotary encoder (GPIO-based)
/ {
    rotary_encoder {
        compatible = "gpio-keys";
        enc_a: enc_a { gpios = <&gpio0 6 (GPIO_ACTIVE_HIGH | GPIO_PULL_UP)>; };
        enc_b: enc_b { gpios = <&gpio0 7 (GPIO_ACTIVE_HIGH | GPIO_PULL_UP)>; };
        enc_sw: enc_sw { gpios = <&gpio0 10 (GPIO_ACTIVE_HIGH | GPIO_PULL_UP)>; };
    };
};

// 4. Buttons
/ {
    buttons {
```

```

compatible = "gpio-keys";
btn1: btn1 { gpios = <&gpio1 11 (GPIO_ACTIVE_LOW | GPIO_PULL_UP)>; };
btn2: btn2 { gpios = <&gpio1 14 (GPIO_ACTIVE_LOW | GPIO_PULL_UP)>; };
btn3: btn3 { gpios = <&gpio0 0 (GPIO_ACTIVE_LOW | GPIO_PULL_UP)>; };
btn4: btn4 { gpios = <&gpio0 1 (GPIO_ACTIVE_LOW | GPIO_PULL_UP)>; };
};
};

```

1.10.2 10.2 Kconfig Additions

```

CONFIG_I2C=y           # already enabled for SHT31
CONFIG_SENSOR=y        # already enabled
CONFIG_BME280=y        # new
CONFIG_SPI=y           # already enabled
CONFIG_SPI_STM32=y      # already enabled
CONFIG_DISPLAY=y        # new (if using Zephyr display API)

```

1.10.3 10.3 No Impact on Existing Functionality

- I2C1 bus: SHT31 at 0x44 + BME280 at 0x76 coexist on same bus
- SPI1/SPI2: HV DACs unaffected (dedicated buses)
- USART3/USART6: console and RS-485 unaffected
- All GPIO bit-bang ADS1232 ADCs unaffected
- SWD debug interface unaffected

1.11 11. Decision Rationale

1. **BME280 on I2C1 (not separate SPI):** Saves 4 pins, uses the already-enabled I2C1 bus. Different I2C address from SHT31 eliminates bus contention. Both sensors have compatible I2C speeds (standard 100 kHz / fast 400 kHz).
2. **SPI3 for SSD1331 (not SPI4/SPI5/SPI6):** SPI3 is the only completely free SPI peripheral where both SCK and MOSI pins are available without conflict. SPI4/SPI5 pins overlap with ADS1232 ADC GPIOs; SPI6 pin overlaps with USART6.
3. **PB cluster for expansion:** All 10 new signals (excluding I2C which is pre-routed) cluster on GPIOB (PB0,PB1,PB3,PB4,PB5,PB6,PB7,PB10,PB11,PB14) + 2 on GPIOC (PC0,PC1). This minimizes expansion PCB trace length and simplifies routing.
4. **No GPIOJ/K:** These ports exist in the STM32F429BI SoC but are not enabled in the jw_hvb board DTS, suggesting they are not broken out to accessible pads/headers.

1.12 12. Open Questions for PCB Designer

- ☐ Are PB0–PB7, PB10, PB11, PB14, PC0, PC1 physically routed to test points, vias, or accessible pads on the jw_hvb PCB?
- ☐ Is VCC_+3V3 accessible at a convenient header or test point near the planned expansion connector location?
- ☐ Does the board have a spare I2C1 bus tap point (other than the SHT31 pads)?
- ☐ What connector pitch and form factor are preferred for the expansion port?
- ☐ Should the expansion sensor PCB have its own 3.3 V LDO (fed from VCC_12V) for noise isolation, or is VCC_+3V3 directly acceptable?